

Experiment No. 1
Explore the descriptive statistics on the given dataset
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Experiment No. 1

1. Aim: Explore the descriptive statistics on the given dataset.

Dataset: In this experiment, fictitious data of Body Mass Index(BMI) containing 10 observations and 5 variables is used. The dataset contains Height, Weight, Age, BMI, and Gender columns.

2. Software used: Google Colaboratory/ Jupyter Notebook

3. Theory :-

Descriptive Statistics:

Descriptive statistics can be defined as the measures that summarize a given data, and these measures can be broken down further

1. Measure of central tendency
2. Measure of spread/dispersion
3. Measure of symmetry/shape

Measure of Central Tendency

Measure of central tendency is used to describe the middle/centre value of the data. Mean, Median, Mode are measures of central tendency.

1. Mean

- Mean is the average value of the dataset.
- Mean is calculated by adding all values in the dataset divided by the number of values in the dataset.
- We can calculate the mean for only numerical variables.

2. Median

- The Median is the middle number in the dataset.
- Median is the best measure when we have outliers.

3. Mode

The mode is used to find the common number in the dataset.

Measure of spread

- The measure of spread/dispersion is used to describe how data is spread. It also describes the **variability** of the dataset.
- **Standard Deviation, Variance, Range, IQR**, are used to describe the measure of spread/dispersion
- The measure of spread can be shown in graphs like **boxplot**.



1. Variance

- Variance is used to describe how far each number in the dataset is from the mean.
- Formula to calculate population variance

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

2. Standard Deviation

- Standard Deviation is the measure of the spread of data from the mean.
- Standard deviation is the square root of variance.
- More the standard deviation, more the spread.

3. Range

- The range is the difference between the largest number and the smallest number.
- Larger the range, the more the dispersion.

4. Interquartile range (IQR)

- Quartiles describe the spread of data by breaking into quarters. The median exactly divides the data into two parts.
- **Q1 (Lower quartile)** is the middle value in the first half of the sorted dataset.
- **Q2**— is the median value
- **Q3 (Upper quartile)** is the middle value in the second half of the sorted dataset
- The interquartile range is the difference between the 75th percentile(Q3) and the 25th percentile(Q1).
- 50% of data fall within this range.

Boxplot is used to describe how the data is distributed in the dataset. This graph represents five-point summary (minimum, maximum, median, lower quartile, and upper quartile) and is used to identify **outliers**.

- whiskers — denote the spread of data
- box— represents the IQR- 50% of data lies within this range.

Measure of shape

1. Skewness

Skewness, which is the measure of the symmetry, or lack of it, for a real-valued random variable about its mean. The skewness value can be positive, negative, or undefined. In a perfectly symmetrical distribution, the mean, the median, and the mode will all have the same value.

2. Kurtosis

Kurtosis provides a measurement about the extremities (i.e. tails) of the distribution of data, and therefore provides an indication of the presence of outliers. Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers.



4. Program:

https://colab.research.google.com/drive/1qLZRk3aTD9kOODw9e9c_ftq71hTJGMha?usp=sharing

Output:

```
→ Mean:
  Height    168.222222
  Weight    73.888889
  bmi       26.177778
  Age       28.333333
  dtype: float64

Median:
  Height    174.0
  Weight    70.0
  bmi       26.1
  Age       28.0
  dtype: float64

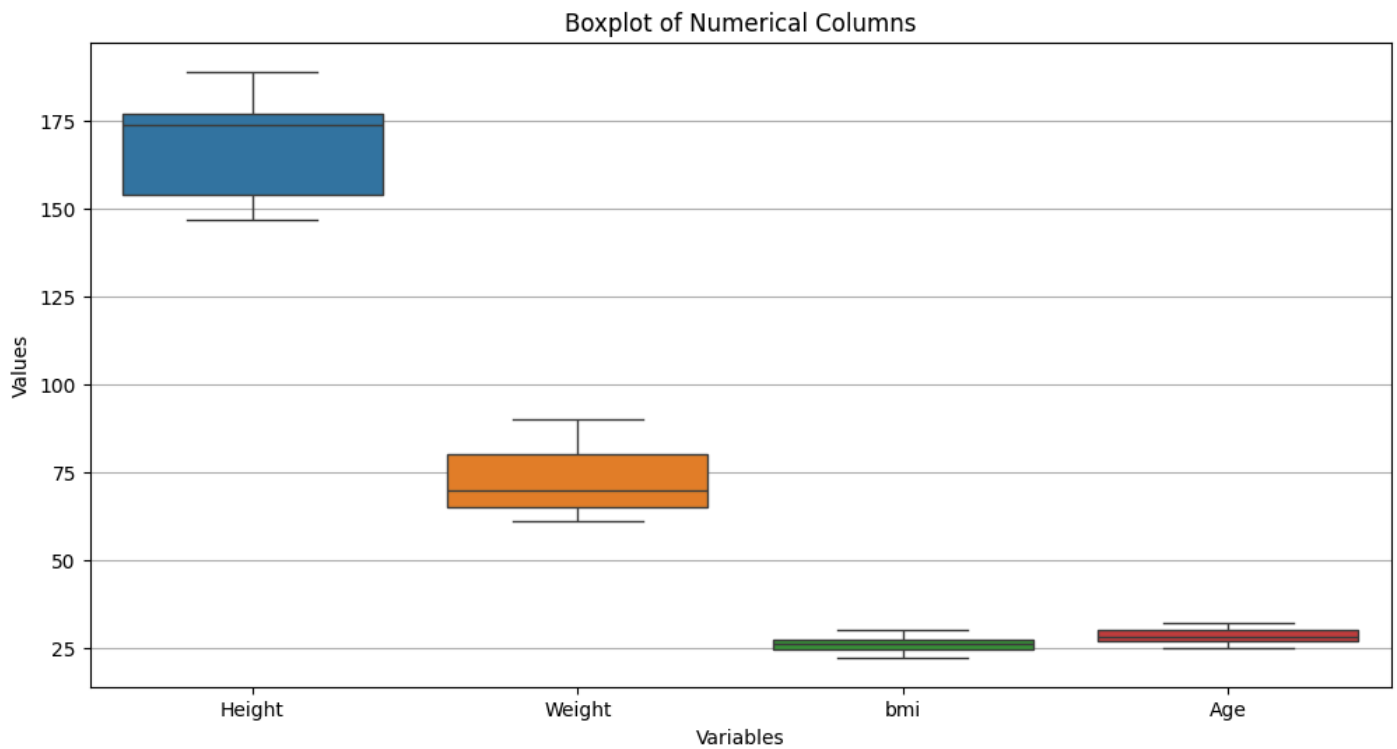
Mode:
{'Height': 174.0, 'Weight': 70.0, 'bmi': 22.3, 'Age': 27.0}
```

```
→ Measures of Spread and Quartiles:

      Q1 (25th Percentile)  Median (Q2, 50th Percentile)  \
Height                    154.0                        174.0
Weight                    65.0                        70.0
bmi                       24.4                        26.1
Age                       27.0                        28.0

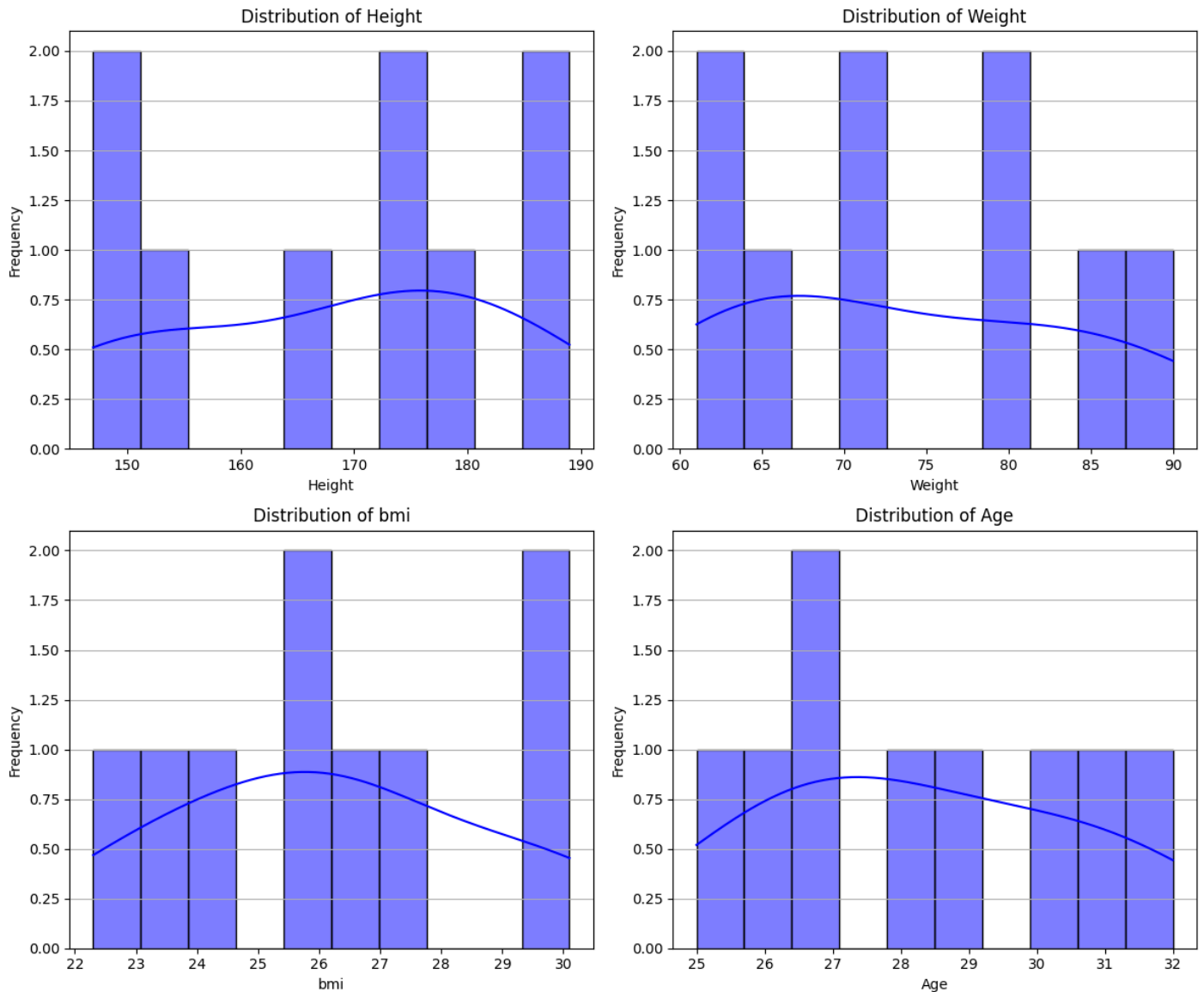
      Q3 (75th Percentile)  IQR (Q3 - Q1)  Variance  Standard Deviation  \
Height                    177.0           23.0  236.194444         15.368619
Weight                    80.0           15.0  115.361111         10.740629
bmi                       27.5            3.1    6.966944          2.639497
Age                       30.0            3.0    5.500000          2.345208

      Range
Height    42.0
Weight    29.0
bmi        7.8
Age        7.0
```



Measures of Shape (Skewness and Kurtosis):

	Skewness	Kurtosis
Height	-0.175894	-1.351014
Weight	0.240826	-1.372808
bmi	0.150582	-1.002889
Age	0.191871	-1.146694



5. Conclusion :- In conclusion, calculating central tendency, dispersion, and shape provides essential insights into the dataset's characteristics. Central tendency helps identify typical values, while dispersion reveals the spread or variability of data. Measures of shape show the distribution's symmetry and skewness. Together, these analyses offer a clearer understanding of trends and consistency within the data. This knowledge is crucial for making informed decisions and guiding further research.